MIT | Academy of Engineering

(An Autonomous Institute Affiliated to Savitribai Phule University)

Name : Pranav Rajendra Patil
PRN : 202501040218
Batch : CS3-C31
Course : Essentials of Data Science (EDS)
Topic : Theory Assignment 2

Dataset Assigned : COVID 19

1) **How many unique countries are in the dataset?**

Problem: Find the total number of unique countries present in the dataset.

Solution:

```
print("Q1 - Total Countries:", df['Country'].nunique())
```

Output: 230

---

2) Top 5 countries with highest Total Cases

Problem: Find the top 5 countries with the most COVID-19 cases.

Solution:

```
print("\nQ2 - Top 5 Countries by Cases:")
print(df[['Country', 'TotalCases']].head(5))
```

Output: USA, India, France, Brazil, Germany

---

3) Country with maximum Total Deaths

Problem: Which country has the highest number of COVID-19 deaths?

Solution:

```
idx = df['TotalDeaths'].idxmax()
print("\nQ3 - Max Deaths:", df['Country'][idx], "-",
int(df['TotalDeaths'][idx]))
```

Output: USA - 1,084,282 deaths

---

4) Average Total Cases across all countries

Problem: Calculate the average number of Total Cases across all countries.

Solution:

```
avg_cases = np.mean(df['TotalCases'].dropna().values)
print("\nQ4 - Average Total Cases:",
round(avg_cases, 2))
```

Output: 2,705,968.93

5) **Countries with more than 1 million Total Cases**

Problem: How many countries have reported more than 1 million Total Cases?

Solution:

```
count = (df['TotalCases'] > 1_000_000).sum()
print("\nQ5 - Countries with 1M+ Cases:", count)
```

Output: 69

---

6) **Minimum COVID Cases Country**

Problem: Find the country with the least number of Total Cases

Solution:

```
idx = df['TotalCases'].idxmin()
print("\nQ6 - Min Cases:", df['Country'][idx], "-",
int(df['TotalCases'][idx]))
```

Output: MS Zaandam - 9 cases

---

7) Total Worldwide Deaths

Problem: Find the sum of all COVID deaths across all countries .

Solution:

```
total_deaths = np.sum(df['TotalDeaths'].dropna().values)
print(total_deaths)
```

Output: 6,547,095 deaths

---

8) Missing Recovered Data

Problem: Find how many countries have missing (NaN) data in TotalRecovered column.

Solution:

```
missing = df['TotalRecovered'].isna().sum()
print(missing)
```

Output: 16 countries

---

9) Highest Tests per 1M Population

Problem: Which country conducted the most tests per 1 million population?

Solution:

```
row = df.loc[df['TestsPer1M'].idxmax()]
print(row['Country'], "-", row['TestsPer1M'])
```

Output: Denmark - 22,004,939 tests/1M

---

10) Standard Deviation of Total Cases

Problem: Calculate the standard deviation of Total Cases to understand the spread.

Solution:

```
std = np.std(df['TotalCases'].dropna().values)
print(std)
```

Output: 8,760,791.59

---

11) Country with highest Active Cases

Problem: Find the country with the highest number of currently Active COVID-19 Cases.

Solution:

```
max_active = df.loc[df['ActiveCases'].idxmax()]
print("Q11 - Highest Active Cases:",
max_active['Country'], "-",
int(max_active['ActiveCases']))
```

Output: USA - 17,349,793

12) Total Recovered cases worldwide

Problem: Find the total number of recovered COVID-19 cases across all countries

Solution:

```
total_recovered =
np.sum(df['TotalRecovered'].dropna().values)
print("\nQ12 - Total Worldwide Recovered:",
int(total_recovered))
```

Output: 580,435,678

---

13) Top 3 countries with highest Deaths per 1M population

Problem: Find the top 3 countries with the most deaths per 1 million population

Solution:

```
print("\nQ13 - Top 3 Deaths per 1M:")
print(df[['Country', 'DeathsPer1M']].nlargest(3,
'DeathsPer1M').to_string(index=False))
```

Output: Peru, Bulgaria, Hungary

---

14) Median of Total Cases

Problem: Find the median value of Total Cases across all countries using NumPy

Solution:

```
median_cases =
np.median(df['TotalCases'].dropna().values)
print("\nQ14 - Median Total Cases:", median_cases)
```

Output: 282,307.5

---

15) Countries where Active Cases are more than Recovered Cases

Problem: Find how many countries have more Active Cases than Total Recovered Cases

Solution:

```
count = (df['ActiveCases'] > df['TotalRecovered']).sum()
print("\nQ15 - Countries with Active > Recovered:", count)
```

Output: 17

---

16) Country with highest Population

Problem: Find the country with the highest population in the dataset.

Solution:

```
max_pop = df.loc[df['Population'].idxmax()]
print("Q16 - Highest Population Country:",
max_pop['Country'], "-",
int(max_pop['Population']))
```

Output: China - 1,439,323,776

---

17) Correlation between Total Cases and Total Deaths

Problem: Find the correlation between Total Cases and Total Deaths using NumPy

Solution:

```
corr = np.corrcoef(df['TotalCases'].dropna(),
df['TotalDeaths'].dropna())[0][1]
print("\nQ17 - Correlation (Cases vs Deaths):",
round(corr, 4))
```

Output: 0.9742 (very strong positive correlation)

---

18) **Sort countries by Total Tests in descending order (Top 5)**
Problem: Sort and display the top 5 countries that conducted the most Total Tests.

Solution:
```
top5_tests = df.nlargest(5, 'TotalTests')[['Country',
'TotalTests']]
print("\nQ18 - Top 5 Countries by Total Tests:")
print(top5_tests.to_string(index=False))
```

Output: USA, India, UK, Russia, Italy

---

19) **Percentage of countries with missing Serious/Critical data**
Problem: Find what percentage of countries have missing data in the SeriousCritical column.

Solution:
```
missing_pct = (df['SeriousCritical'].isna().sum() /
len(df)) * 100
print("\nQ19 - % Countries with Missing Serious/Critical
Data:",
round(missing_pct, 2), "%")
```

Output: 36.52%

---

20) Minimum Deaths per 1M population (excluding zeros)

Problem: Find the country with the lowest Deaths per 1M population (excluding 0 values).

Solution:

```
filtered = df[df['DeathsPer1M'] > 0] min_row =
filtered.loc[filtered['DeathsPer1M'].idxmin()]
print("\nQ20 - Lowest Deaths per 1M:",
min_row['Country'], "-", min_row['DeathsPer1M'])
```

Output: North Korea - 0.07
